

AARSH PATEL

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EXPERIENCE

Codametrix

March 2024 – March 2026

Applied Research Scientist

Boston, MA

- Improved ICD and CPT medical code classification and extraction F1 by 20% by replacing SVM-based models with fine-tuned BERT/LLM components within the medical coding platform.
- Developed and integrated NLP-based information extraction pipelines from clinical notes and discharge summaries, improving end-to-end automation of medical case processing by 15% across multiple health systems.
- Built a confidence threshold calibration framework enabling product and clinical teams to meet health system-defined quality targets (e.g., 95% acceptance rate), translating SLA requirements into ML pipeline configurations to maximize automation rates.
- Built and deployed a primary ICD code verifier using a fine-tuned Llama model, achieving 96% F1 on primary code placement by correctly reordering ICD predictions from the upstream coding pipeline.

Orbita

April 2022 – December 2023

Senior Machine Learning Engineer

Boston, MA

- Drove adoption across 10+ healthcare organizations by leading end-to-end development and deployment of a patient-facing Generative AI Virtual Assistant that reduces inbound call volume.
- Built the core RAG and prompt engineering framework powering the virtual assistant, leveraging OpenAI GPT models to deliver accurate, context-aware responses at production scale.
- Designed and implemented Blaze, an NLP-based entity extraction framework for medical terms, directly improving bot decision-making accuracy and clinical relevance of assistant responses.

Amazon (Alexa)

June 2020 – November 2021

Applied Research Scientist

Cambridge, MA

- Reduced annotation costs significantly and cut model training time by 20% by applying semi-supervised learning over millions of unlabeled utterances to build production-scale NLU models for intent and domain classification.
- Matched fully supervised baseline NLU performance using only 50% of labeled data, demonstrating a scalable and cost-efficient approach to model training.
- Leveraged LLM (BERT, GPT) embeddings and unsupervised clustering to build scalable pseudo-labeling pipelines, automating annotation across millions of unlabeled utterances.

Amazon (Alexa)

September 2019 – December 2019

Applied Research Scientist Intern

Cambridge, MA

- Conducted research on fairness in NLU systems for Alexa, proposing a novel approach combining BERT embeddings and unsupervised cohort discovery to reduce performance variance by 10–12%, resulting in a co-authored workshop paper accepted at ICLR 2020.

EDUCATION

University of Massachusetts Amherst

January 2018 – May 2020

Master of Science in Computer Science

Amherst, MA

University of Massachusetts Amherst

September 2014 – December 2017

Bachelor of Science in Computer Science

Amherst, MA

SKILLS

Languages: Python, JavaScript, SQL

Tools & Frameworks: Databricks, Docker, Git, Jira, AWS, Azure, React JS, Node JS, LaTeX, PyTorch, Keras, scikit-learn, Matplotlib, Pandas, LangChain, Pinecone VectorDB, Hugging Face

PUBLICATIONS

Towards Classification Parity Across Cohorts

December 2019

Accepted at ML Workshop, ICLR 2020

Research on algorithmic fairness that expands beyond explicit demographic variables by defining implicit cohorts — groups sharing similar language patterns discovered through utterance embedding clustering — and systematically evaluates classification performance gaps across both cohort types.